

§3-10 Related Rates

Homework : 3,5,7,8,9,12,14,17,19,25,31,35,38

Related Rates : 若問題出現如下敘述 :

討論某些量 x 對時間 t 的變化率 (rate of change) 的應用問題.

at what rate x? } 則是問 $\frac{dx}{dt} = ?$ (即 x 對時間 t 的變化率).
How fast(slow)..... x? }

Steps to solve related rates problems.

1. 引進“符號”將題目(文字)轉成“圖”.
2. 找一方程式“連結”此圖.
3. $\frac{d}{dt}$ (此方程式).

Example1 :

Air is being pumped into a spherical balloon so that its volume increases at a rate of $100 \text{ cm}^3/\text{s}$. How fast is the radius of the balloon increasing when the diameter is 50 cm?

Solution : 已知 $\frac{dV}{dt} = 100$, 求 $\frac{dr}{dt}|_{r=25}$.

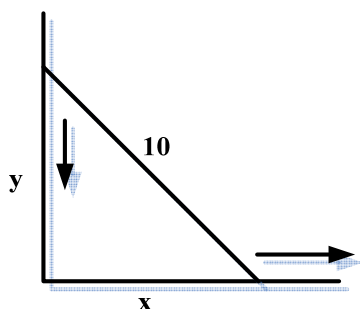
$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{1}{25\pi} (\text{cm}^3/\text{s}).$$

Example2 :

A ladder 10 ft long rests against a vertical wall. If the bottom of the ladder slides away from the wall at a rate of 1 ft/s. How fast is the top of the ladder sliding down the wall when the ladder is 6 ft from the wall?

Solution :



已知: $\frac{dx}{dt} = 1$ (ft/s)

求: $\frac{dy}{dt} \Big|_{x=6}$.

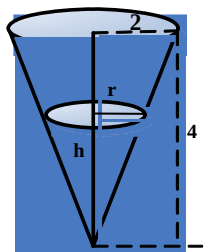
$$x^2 + y^2 = 100 \Rightarrow 2x \left(\frac{dx}{dt} \right) + 2y \left(\frac{dy}{dt} \right) = 0. \text{ 當 } x = 6 \Rightarrow y = 8.$$

$$\Rightarrow \frac{dy}{dt} = -\frac{3}{4} \text{ (ft/s)}.$$

Example3 :

A water tank has the shape of an inverted circular cone with base radius 2 m and height 4 m. If water is being pumped into the tank at a rate of $2 \text{ m}^3/\text{min}$, find the rate at which the water level is rising when the water is 3 m deep.

Solution :



已知: $\frac{dV}{dt} = 2$ (m^3/min).

求: $\frac{dh}{dt} \Big|_{h=3} = ?$

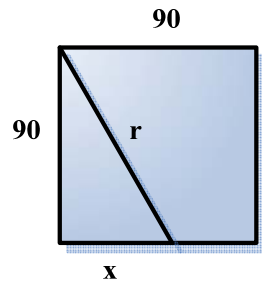
$$V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi \left(\frac{h}{2} \right)^2 h = \frac{\pi}{12} h^3$$

$$\Rightarrow \frac{dV}{dt} = \frac{\pi}{4} h^2 \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{8}{9\pi} (\text{m}^3/\text{min}).$$

Example4 :

A baseball diamond is a square with side 90 ft. A batter hits the ball and runs toward first base with a speed of 24 ft/s. At what rate is his distance from third base increasing when he is half way to first base?

Solution :



已知: $\frac{dx}{dt} = 24$ (ft/s).

求 : $\frac{dr}{dt} \Big|_{x=45} = ?$

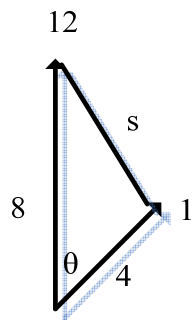
$$r^2 = 90^2 + x^2 \Rightarrow 2r \frac{dr}{dt} = 2x \frac{dx}{dt}$$

$$\Rightarrow \frac{dr}{dt} = \frac{x}{r} \frac{dx}{dt} = \frac{24}{\sqrt{5}} = \frac{24\sqrt{5}}{5}. (\because \text{當 } x = 45, r = 45\sqrt{5}.)$$

Example5 :

The minute hand on a watch is 8 mm long and the hour hand is 4 mm long. How fast is the distance between the tips of the hands changing at one o'clock?

Solution :



已知: $\frac{d\theta}{dt} = \frac{\pi}{6} - 2\pi = -\frac{11}{6}\pi$ (rad/h)

分針的角速度 = 2π

時針的角速度 = $\frac{\pi}{6}$

求 : $\frac{ds}{dt} \Big|_{\theta=\frac{\pi}{6}} = ?$

$$s^2 = 64 + 16 - 64 \cos \theta$$

$$\Rightarrow 2s \frac{ds}{dt} = 64 \sin \theta \frac{d\theta}{dt} \Rightarrow \frac{ds}{dt} = \frac{-88\pi}{3\sqrt{80-32\sqrt{3}}} \text{ (rad/h)}.$$

(When $\theta = \frac{\pi}{6}$, $s = \sqrt{80-32\sqrt{3}}$).